Accept-Reject Methods

Monte Carlo Statistical Methods

Sai Sampath Kedari
sampath@umich.edu

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Contents

1 Accept–Reject: Intuition, Construction, and Proof

1.1 Setup and Goal

We want to generate random samples from a target density $f(x)$ on a support $\mathcal{X}$, but direct sampling is hard.

We choose a proposal density $g(x)$ on the same support $\mathcal{X}$, easy to sample from, and a constant $M \geq 1$ such that

$$f(x) \leq M\, g(x) \quad \text{for all } x \in \mathcal{X}.$$

Geometrically, the curve $Mg(x)$ covers $f(x)$ everywhere.

1.2 Fundamental geometric idea

If we could sample uniformly under the curve of $f(x)$,

$$(X, U) \sim \text{Uniform}\big\{(x, u) : 0 < u < f(x)\big\},$$

then the x–coordinates alone would follow $f(x)$. This is the core idea we will recreate indirectly using an easier region under $Mg(x)$.

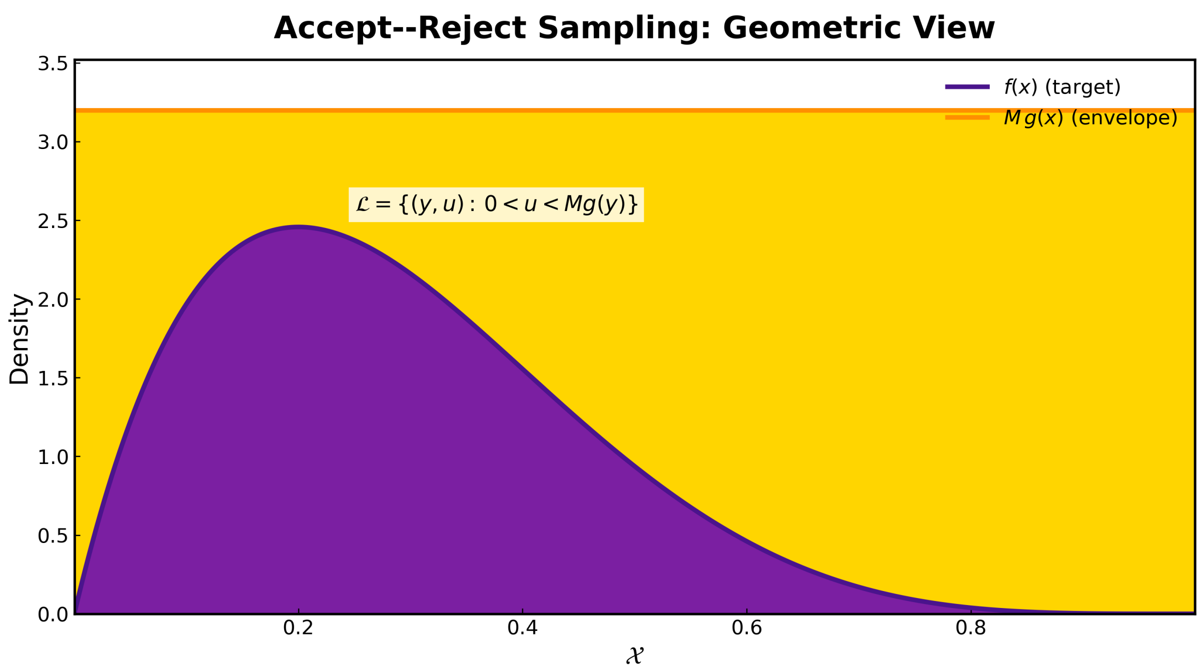

Figure 1: Geometric illustration of the Accept–Reject method. The total region $\mathcal{L} = \{(y, u) : 0 < u < Mg(y)\}$ is shown in yellow. The accepted region under $f(y)$ is filled in purple, representing the set $\{(y, u) : 0 < u < f(y)\}$. By projecting the accepted points onto the y–axis (domain $\mathcal{X}$), we obtain samples distributed according to the target density $f(y)$.

1.3 Joint construction on a 2D region

Define the 2D support set

$$\mathcal{L} \;=\; \big\{(y, u) : 0 < u < Mg(y)\big\}.$$

We generate a pair (Y, U) uniformly on $\mathcal{L}$ by sampling

$$Y \sim g, \qquad U \mid Y = y \sim \text{Uniform}\big(0,\, Mg(y)\big).$$

Equivalently, the joint density of (Y, U) is

$$p_{Y,U}(y,u) = \begin{cases} \dfrac{1}{M}, & 0 < u < Mg(y), \\ 0, & \text{otherwise.} \end{cases}$$

Now define the acceptance rule

$$\text{accept if } U < f(Y), \qquad \text{and set } X := Y.$$

Intuition: we are sampling uniformly under the envelope Mg, then keeping only the points that lie below f. The accepted points are uniform under the region $\{(y,u) : 0 < u < f(y)\}$, so their y–coordinates have density f.

1.4 Proof that accepted samples follow f

Let $A \subseteq \mathcal{X}$ be any measurable set. By the acceptance rule,

$$P(X \in A) = P\big(Y \in A \mid U < f(Y)\big) = \frac{P\big(Y \in A,\, U < f(Y)\big)}{P\big(U < f(Y)\big)}.$$

Compute numerator and denominator using the uniform density $1/M$ on $\mathcal{L}$:

$$P\big(Y \in A,\, U < f(Y)\big) = \int_A \int_0^{f(y)} \frac{1}{M}\, du\, dy = \frac{1}{M} \int_A f(y)\, dy,$$

$$P\big(U < f(Y)\big) = \int_{\mathcal{X}} \int_0^{f(y)} \frac{1}{M}\, du\, dy = \frac{1}{M} \int_{\mathcal{X}} f(y)\, dy = \frac{1}{M}.$$

Therefore,

$$P(X \in A) = \frac{\frac{1}{M} \int_A f(y)\, dy}{\frac{1}{M}} = \int_A f(y)\, dy.$$

Since this holds for all measurable A, the accepted X has density f.

1.5 Acceptance probability and efficiency

We now compute the probability that a proposed sample is accepted. From the joint construction on $\mathcal{L}$, we have

$$p_{Y,U}(y,u) = \begin{cases} \dfrac{1}{M}, & 0 < u < Mg(y), \\ 0, & \text{otherwise.} \end{cases}$$

Therefore,

$$\begin{aligned}
P(\text{accept}) &= P\big(U < f(Y)\big) \\
&= \int_{\mathcal{X}} \int_0^{f(y)} \frac{1}{M}\, du\, dy \\
&= \frac{1}{M} \int_{\mathcal{X}} f(y)\, dy \\
&= \frac{1}{M},
\end{aligned}$$

since f is a valid probability density function.

Hence, the number of proposals T required until the first acceptance follows a geometric distribution with success probability $1/M$, and

$$\mathbb{E}[T] = \frac{1}{P(\text{accept})} = M.$$

This shows that on average, we must draw M proposals to obtain one accepted sample.

Geometric intuition (area ratio). Because (Y, U) is sampled uniformly on the region

$$\mathcal{L} = \{(y, u) : 0 < u < Mg(y)\},$$

the acceptance probability can also be viewed as the ratio of two areas:

$$P(\text{accept}) = \frac{\text{Area under } f}{\text{Area under } Mg} = \frac{\int_{\mathcal{X}} f(y)\, dy}{\int_{\mathcal{X}} Mg(y)\, dy} = \frac{1}{M}.$$

Hence, the acceptance rate equals the fraction of the total area of $\mathcal{L}$ occupied by the region under f, providing a simple geometric interpretation of efficiency.

Unnormalized targets. If the target density is only known up to a normalizing constant, say

$$f(x) = \frac{h(x)}{Z}, \quad Z = \int_{\mathcal{X}} h(y)\, dy,$$

and we use the acceptance test

$$U < \frac{h(Y)}{Mg(Y)},$$

then by the same reasoning,

$$\begin{aligned}
P(\text{accept}) &= \int_{\mathcal{X}} \int_0^{h(y)} \frac{1}{MZ}\, du\, dy \\
&= \frac{1}{MZ} \int_{\mathcal{X}} h(y)\, dy \\
&= \frac{1}{M}.
\end{aligned}$$

Thus, the acceptance rate remains $\frac{1}{M}$; the unknown normalization constant cancels out.

1.6 Algorithm (one line per step)

1) Draw $Y \sim g, \quad U \sim \text{Uniform}(0, 1)$.

2) If $U < \dfrac{f(Y)}{Mg(Y)}$, set $X := Y$ (accept).

3) Otherwise reject and repeat.

1.7 Why this is intuitive

We simulate uniformly under the easy surface Mg and keep only the portion that lies under f. The accepted cloud is uniform under the region below f, so its x–projection has the exact shape of f. This is the geometric heart of Accept–Reject.

2 Sampling $\mathcal{N}(0, 1)$ Using a Laplace Proposal (Accept–Reject Method)

Problem Setup

We want to generate samples from the target density

$$f(x) = \frac{1}{\sqrt{2\pi}}\, e^{-x^2/2}, \qquad x \in \mathbb{R},$$

using the Accept–Reject method.

As the proposal, we choose the symmetric **Laplace** (double-exponential) distribution:

$$g(x) = \frac{1}{2b}\, e^{-|x|/b}, \qquad x \in \mathbb{R},$$

where $b > 0$ is the scale parameter. The Laplace proposal is easy to sample from and has exponential tails that provide a good envelope over the Gaussian target.

2.1 Finding the Envelope Constant M

To apply the Accept–Reject method, we must find the smallest constant M such that

$$f(x) \le Mg(x) \quad \text{for all } x \in \mathbb{R}.$$

This leads to

$$\begin{aligned}
M &= \sup_{x\in\mathbb{R}} \frac{f(x)}{g(x)} \\
&= \sup_{x\in\mathbb{R}} \frac{\frac{1}{\sqrt{2\pi}} e^{-x^2/2}}{\frac{1}{2b} e^{-|x|/b}} \\
&= \frac{2b}{\sqrt{2\pi}} \sup_{x\ge 0} \exp\left(-\frac{x^2}{2} + \frac{x}{b}\right),
\end{aligned}$$

where we use symmetry to restrict to $x \ge 0$.

Maximizing the Exponent

To find the maximum of the exponential term, differentiate the exponent with respect to x:

$$\frac{d}{dx}\left(-\frac{x^2}{2}+\frac{x}{b}\right)=-x+\frac{1}{b}.$$

Setting this derivative equal to zero gives

$$x^*=\frac{1}{b}.$$

Evaluating the exponent at this point,

$$-\frac{(x^*)^2}{2}+\frac{x^*}{b}=-\frac{1}{2b^2}+\frac{1}{b^2}=\frac{1}{2b^2}.$$

Substituting back, we obtain

$$M=\frac{2b}{\sqrt{2\pi}}\exp\left(\frac{1}{2b^2}\right).$$

Optimizing over b

If the scale parameter b is adjustable, we can minimize $M(b)$ with respect to b to find the most efficient proposal. Differentiating and setting $\frac{dM}{db}=0$ gives

$$\begin{aligned}
M(b)&=\frac{2b}{\sqrt{2\pi}}\,e^{1/(2b^2)},\\
\frac{dM}{db}&=\frac{2}{\sqrt{2\pi}}\,e^{1/(2b^2)}\left(1-\frac{1}{b^2}\right)=0.
\end{aligned}$$

Hence, the minimum occurs at

$$b^*=1.$$

Substituting this value yields

$$M^*=\frac{2}{\sqrt{2\pi}}\,e^{1/2}\approx 1.3154.$$

Interpretation

The optimal proposal is therefore the Laplace$(0,1)$ distribution, with an envelope constant of approximately $M=1.316$. This means the Laplace proposal slightly overestimates the Gaussian density, accepting about

$$\frac{1}{M}\approx 0.76$$

of proposed samples on average. The close match between the Gaussian and Laplace shapes shows that the Laplace envelope provides an efficient, near-optimal cover for $\mathcal{N}(0,1)$.

2.2 Summary of Parameters

$$\text{Target:} \quad f(x) = \frac{1}{\sqrt{2\pi}}\, e^{-x^2/2},$$
$$\text{Proposal:} \quad g(x) = \frac{1}{2}\, e^{-|x|},$$
$$\text{Envelope Constant:} \quad M = \frac{2}{\sqrt{2\pi}}\, e^{1/2} \approx 1.316.$$

Next Step: Simulation

We now implement the Accept–Reject algorithm using this configuration:

$$f(x) = \mathcal{N}(0,1), \qquad g(x) = \mathrm{Laplace}(0,1), \qquad M = 1.316,$$

and empirically verify that the accepted samples follow the $\mathcal{N}(0,1)$ distribution.

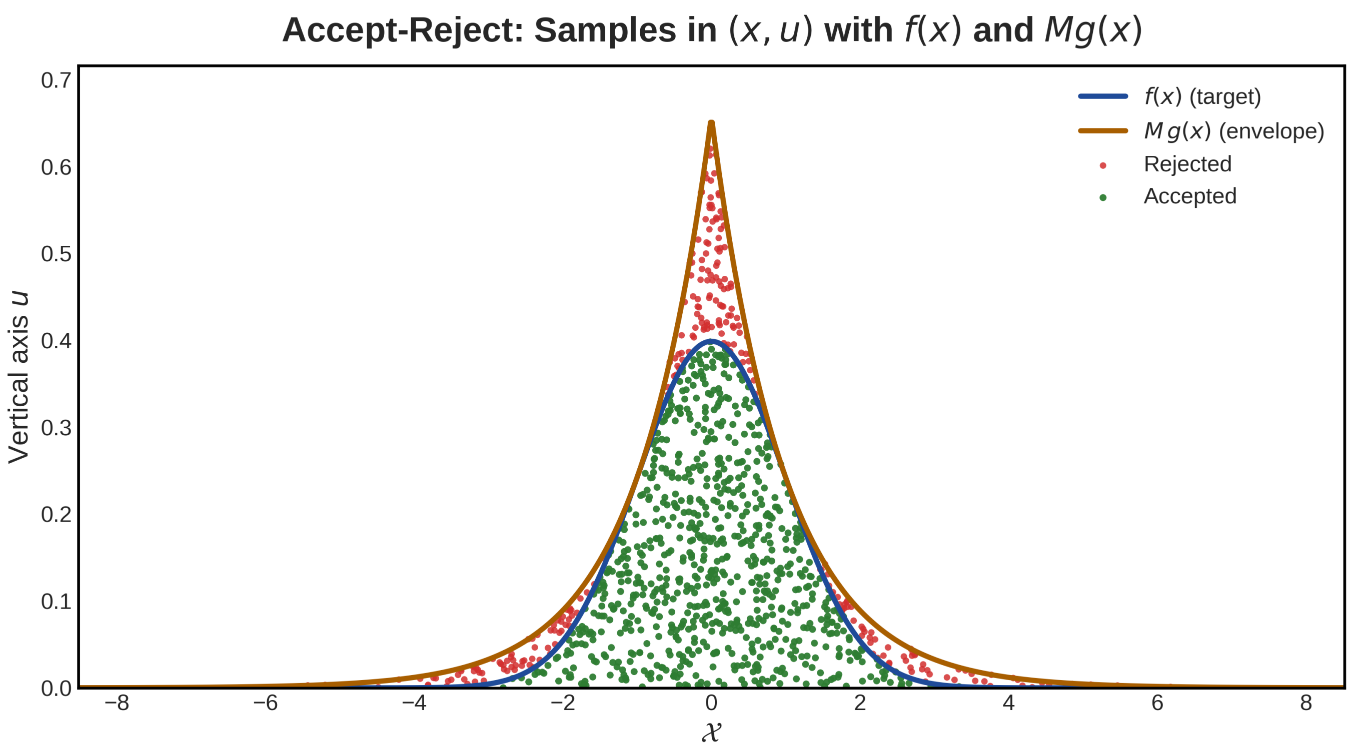

Figure 2: Accept–Reject visualization for sampling $\mathcal{N}(0,1)$ using a Laplace(0, 1) proposal. The blue curve represents the target $f(x)$, the brown curve represents the envelope $Mg(x)$, green dots indicate accepted samples, and red dots indicate rejected proposals. Approximately 76 of proposals are accepted, consistent with the theoretical efficiency $\frac{1}{M} \approx 0.76$

Diagnostics

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Diagnostics:
  total_proposals       : 1053.00000
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acceptance_rate      : 0.75973
expected_acceptance  : 0.75988
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